

Why Customers Leave

Root-cause & causal evidence brief · three independent lines of evidence on 93,358 customers and 11,424 low-score reviews

74.1%	+3.9 pp	−0.46	81.3%
Dormant (180 d caliber)	Late delivery → churn (PSM ATT)	Late delivery → satisfaction	Customers whose TOP-1 driver is delivery

“Customers are leaving” is a symptom. This brief answers the diagnostic question — **why** — and then separates correlation from cause, because the retention budget should follow the causal driver, not the loudest one.

1 · Three independent lines of evidence

Line of evidence	Method	Question it settles
Voice of the customer	LLM-assisted NLP over 11,424 low-score reviews (300 sampled, ±5 % at 95 %)	What do customers say is wrong?
Operational dose-response	delay-day buckets vs low-score rate	How much delay is too much?
Causal identification	PSM · IV/2SLS · Doubly-Robust (DML) · DAG d-separation	How much of the churn does delivery actually cause?

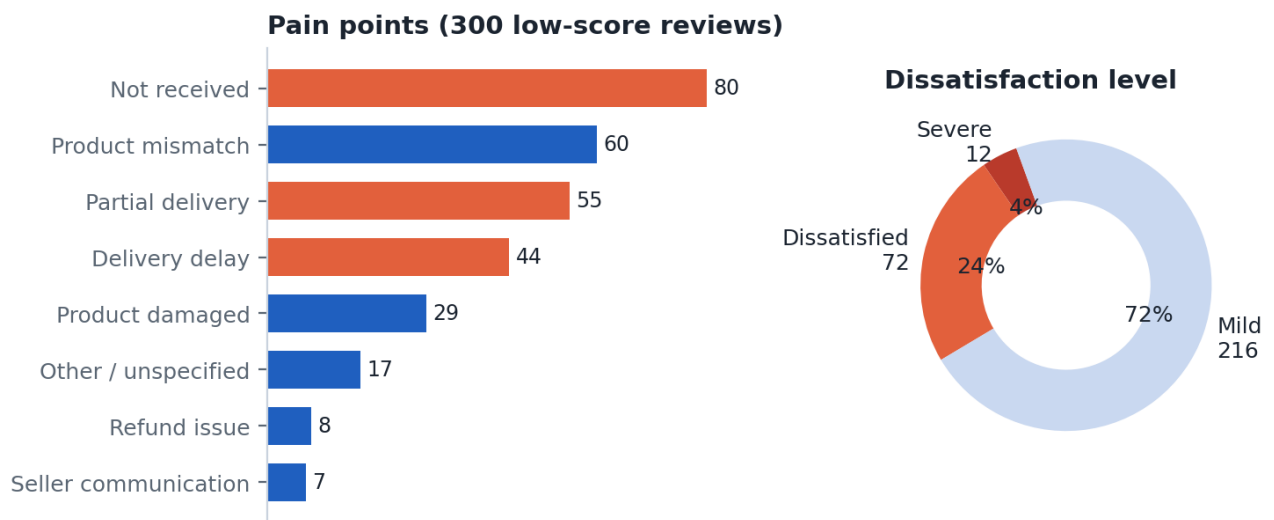
The three agree. That agreement — not any single p-value — is why the recommendation is delivery reliability rather than, say, discounting.

2 · Voice of the customer — what the reviews actually say

11,424 reviews scored ≤ 2 (of ~99 K). A 300-review stratified sample was translated from Portuguese, classified into an 8-category pain-point taxonomy with sub-types and sentiment intensity. Top four pain points account for 79.7 % of complaints; top five for 89.4 %.

Rank	Pain point	Share	Mean sentiment	Reading
1	not_received	26.7 %	0.85	the order never arrived — the hardest failure
2	product_mismatch	20.0 %	0.82	item differs from the listing
3	partial_delivery	18.3 %	0.82	incomplete parcel — warehouse picking
4	delivery_delay	14.7 %	0.79	arrived, but far too late
5	product_damaged	9.7 %	0.83	packaging / QC

Two readings matter for operations: among the angriest reviews (score = 1) not_received rises to 29.8 % versus 11.5 % at score = 2 — non-delivery is the bottom-line breaker; and 74 % of all low-score reviews carry sentiment intensity ≥ 0.7, meaning there is no such thing as a mild complaint in this dataset. The Portuguese original “Não recebi meu pedido” is the single most frequent phrase.

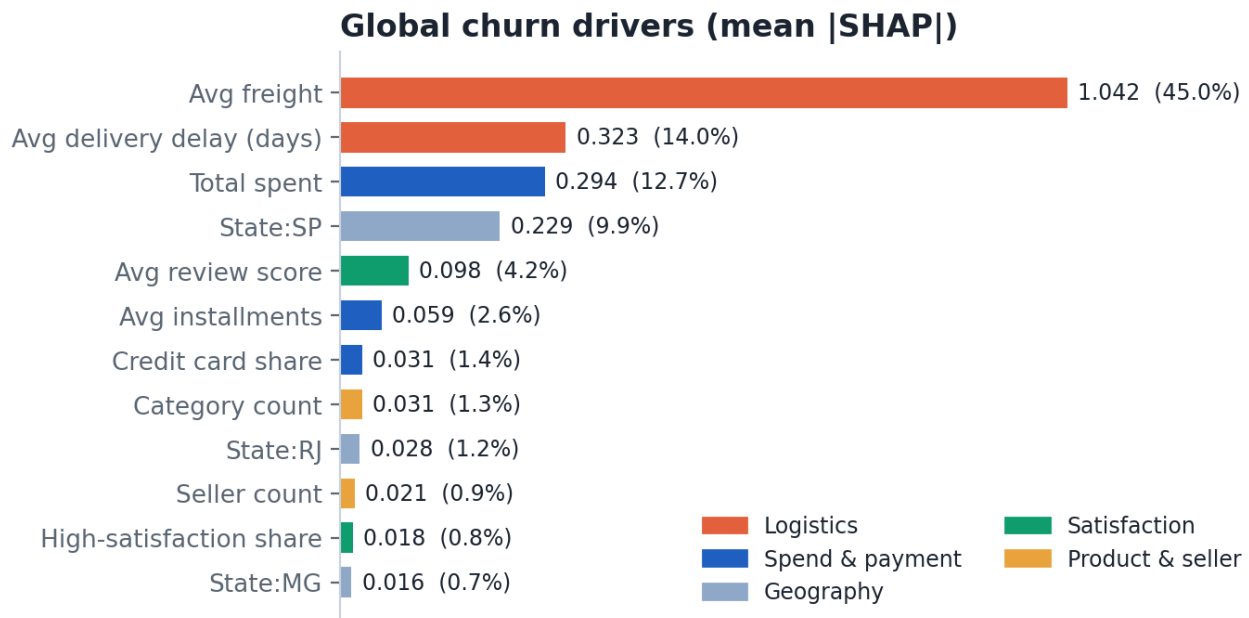


Review pain-point mix — the four delivery-related categories dominate the complaint base.

3 · Operational dose-response — how much delay is too much

Delay days are bucketed and plotted against the share of low-score reviews. The relationship is not linear: it inflects hard in the **3–7 day** band, where the low-score rate reaches **62 %**. Beyond that the curve flattens — customers have already given up, which is why late recovery campaigns underperform early ones.

The caliber warning attached to this finding is as important as the finding itself: only **155 orders** out of ~97 K arrived exactly on the promised date, against 88,552 early deliveries. “On time” is therefore a statistical artefact, not an operating target. The target is the **±1 day window**.



Driver ranking from the predictive layer, shown here because it independently lands on the same two drivers as the NLP layer and the causal layer.

4 · Causal identification — separating driver from coincidence

Late delivery is correlated with churn. It is also correlated with distance, with low-value baskets and with difficult categories. Four estimators were run to see whether the effect survives adjustment:

Estimator	Target	Estimate	Verdict
PSM (propensity-score matching)	churn	+3.9 pp	significant; raw gap 6.4 pp, so ~40 % of the raw gap is selection
PSM (propensity-score matching)	satisfaction	−0.46 points	significant; raw gap −1.72
IV / 2SLS	repurchase	not significant (p = 0.61)	repurchase chain stays unidentified
Doubly-Robust / DML	repurchase	≈ 0 effect	agrees with IV
DAG d-separation	identification set	{OrderCount, CustomerState}	confounder set made explicit and testable

The honest headline: lateness demonstrably raises churn and demonstrably lowers satisfaction. It does **not** demonstrably move repurchase in this dataset — partly because repurchase is only 2.8 % of the base, so the event count is too thin to identify. The recommendation is made on the churn and satisfaction effects, and the deck says so explicitly.

5 · Targeting — who to act on first

A T-Learner CATE model ranks customers by expected uplift rather than by risk. The direction is usable; the ranking is not. In the top decile there are 12 treated repurchases against 234 controls — a 20 : 1 event imbalance

that makes rank ordering unstable ($AUUC \approx 0$). The report therefore recommends targeting by the **stable causal segment** instead: customers with a single historical order and a clean delivery record, where the estimated uplift is largest, rather than by a decile that a re-run would reshuffle.

6 · What did not work — reported, not hidden

- **The repurchase causal chain is unidentified.** Under all four estimators the effect collapses to zero (IV $p = 0.61$). The 20:1 imbalance is the reason, and it is stated on the dashboard, not buried.
- **Customer-service-led retention does not pay.** Modelled ROI $0.78\times$ — below break-even. It is in the strategy table next to the winners precisely so the comparison is visible.
- **Uplift ranking is unreliable.** The model is kept for the direction of the effect only.
- **“Repeat buyers spend $2.01\times$ ” is flattering.** After removing same-day split orders the multiple is $1.55\times$, and 29 % of repeat buyers are single-session splits.

7 · What this changes

Delivery reliability is the one lever that is simultaneously the largest reported driver (81.3 % of customers), the largest causal effect on churn, and the cheapest to instrument. Logistics improvement returns an estimated **$4.34\times$** its cost, against $4.01\times$ for product listing QA and $2.21\times$ for the combined programme. The recommendation to the client is therefore: instrument the ± 1 day window first, act on the high-risk list weekly, and re-measure the repurchase effect on a larger cohort before spending against it.

Every figure in this brief is reproducible from the project repository — SQL in `sql/`, analysis in `scripts/`, and the monitoring view in the Power BI dashboard.

How this was produced. Human-led method, AI-assisted execution. The business problem, the analysis framework, the choice of method and the validation of every result are mine; AI handled implementation and drafting. Each figure is reconciled by an automated gate before it ships, and no claim reaches a client document without a human having checked it against its source snapshot.